

$$\int_0^1 \frac{x(2x^2+1)}{\sqrt{x^4+x^2+1}}dx$$

$$u:=2x^2+1 \quad du=4x$$

$$2x^2=u-1 \Longleftrightarrow x^2=\frac{u-1}{2}$$

$$\frac{1}{4}\int_1^3\frac{u}{\sqrt{\frac{(u-1)^2}{4}+\frac{u-1}{2}+1}}du$$

$$\frac{1}{4}\int_1^3\frac{u}{\sqrt{\frac{1}{4}(u^2-2u+1)+2(u-1)+4}}du$$

$$\frac{1}{2}\int_1^3\frac{u}{\sqrt{u^2-2u+2u-2+5}}du$$

$$\frac{1}{2}\int_1^3\frac{u}{\sqrt{u^2+3}}du$$

$$w:=u^2+3 \quad dw=2udu$$

$$\frac{1}{4}\int_4^{12}\frac{1}{\sqrt{w}}dw$$

$$\frac{1}{2}\left(w^{\frac{1}{2}}\right)|_4^{12}$$

$$\frac{1}{2}\left(\sqrt{12}-\sqrt{4}\right)|_4^{12}$$

$$\frac{1}{2}\left(2\sqrt{3}-2\right)$$

$$\sqrt{3}-1$$